Can we predict Falcon 9 first-stage recovery?

Historical launches • Data cleaning • EDA • SQL • Maps • Predictive modeling

90	60	3	2010–2020
Launch records	Positive outcomes	Launch sites	Time window

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Falcon 9 landing outcome | historic teaching sample

Executive summary

01 Business question

Estimate historical first-stage landing outcomes to support competitive launch intelligence.

02 Analytical approach

Structured API-derived records; data cleaning; grouping/SQL; site and orbit visuals; map/dashboard workflows.

03 Data-backed finding

60/90 observations have a positive recovery label (66.7%). Rates vary by site and year.

04 Limitation

This deck uses a public teaching snapshot, not a newly collected or independently verified live feed.

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Introduction / business problem

SpaceX reduces launch costs in part by recovering and reusing Falcon 9 first stages. A prospective competitor benefits from understanding when recovery is more likely, but historic associations are not a cost quote or a guarantee of future outcomes.

Prediction target

Class = 1 denotes a positive first-stage recovery outcome in this teaching dataset.

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Project methodology

1. **SpaceX API**
   Retrieve launches, boosters, pads, payload details
2. **Wikipedia**
   Scrape historical launch table using pandas / BeautifulSoup
3. **Wrangling**
   Check nulls, payload imputation, labels, categorical features
4. **Analysis**
   Python plots, SQL aggregations, Folium map, Dash
5. **Modeling**
   Train/test split; LR, SVM, tree and KNN; tune and evaluate

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Data collection | SpaceX API

01 GET /v4/launches/past → launch records

02 Join rocket / launchpad / payload / booster identifiers

03 Normalize nested JSON into one row per launch

04 Store launch date, orbit, site, payload and recovery fields

Methodology reflects the IBM capstone workflow. Results in this deck are from a public archived prepared dataset; new API calls were not run for this report.

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Data collection | Wikipedia scraping

01 Open historical Falcon 9 / Falcon Heavy launches page

02 Parse launch tables and table headers

03 Standardize dates, payload mass, mission, and outcomes

04 Export an auditable CSV; cross-check against API-derived launches

Web-scraping is shown as an assignment methodology; the current numeric analyses use the publicly available prepared SpaceX teaching CSV.

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Data wrangling / feature engineering

Quality controls

Parse Date; inspect missing values and types; flag records requiring imputation.

Numeric and binary

PayloadMass, Flights, ReusedCount; GridFins, Reused, Legs.

Categorical encoding

One-hot encode Orbit and LaunchSite for estimators.

Leakage prevention

Fit transformations on training folds only; avoid using Outcome text as a feature.

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Exploratory data analysis | launch trend

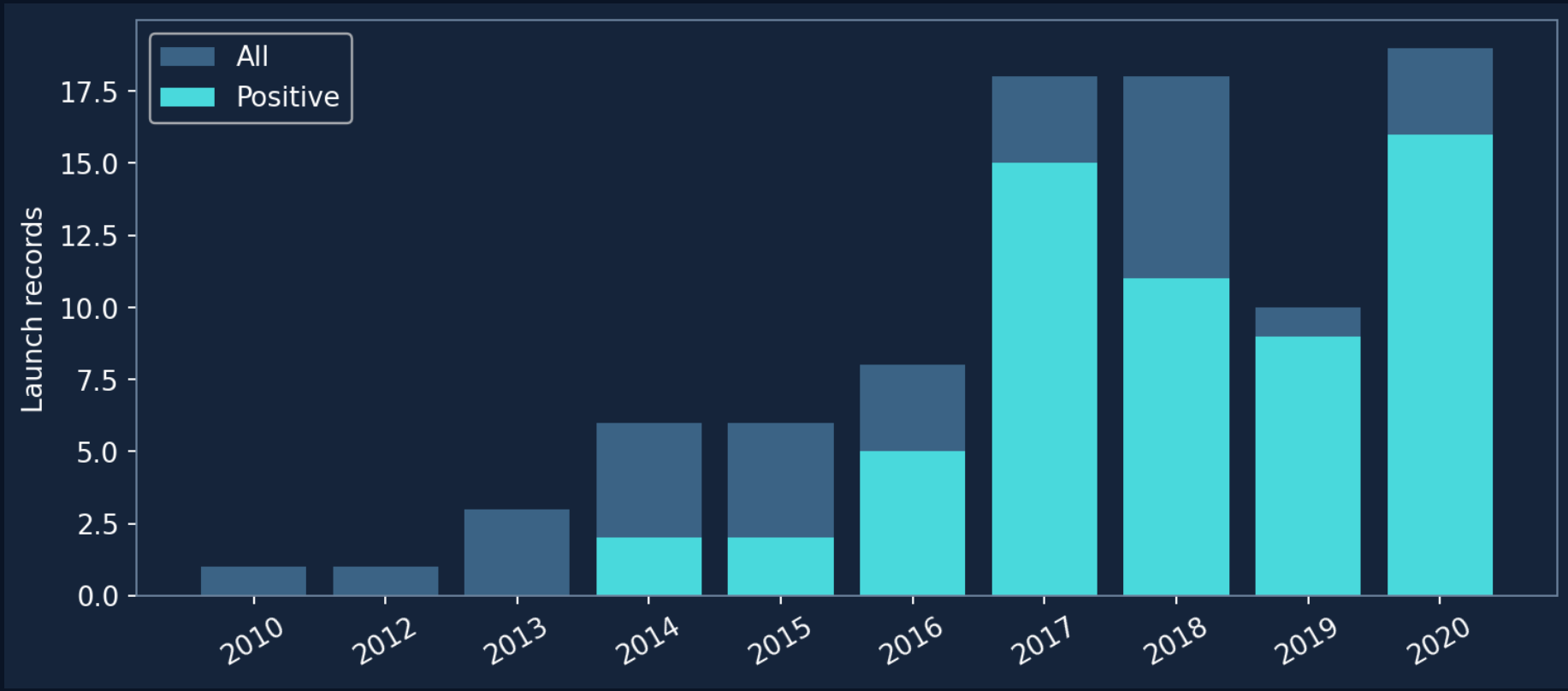
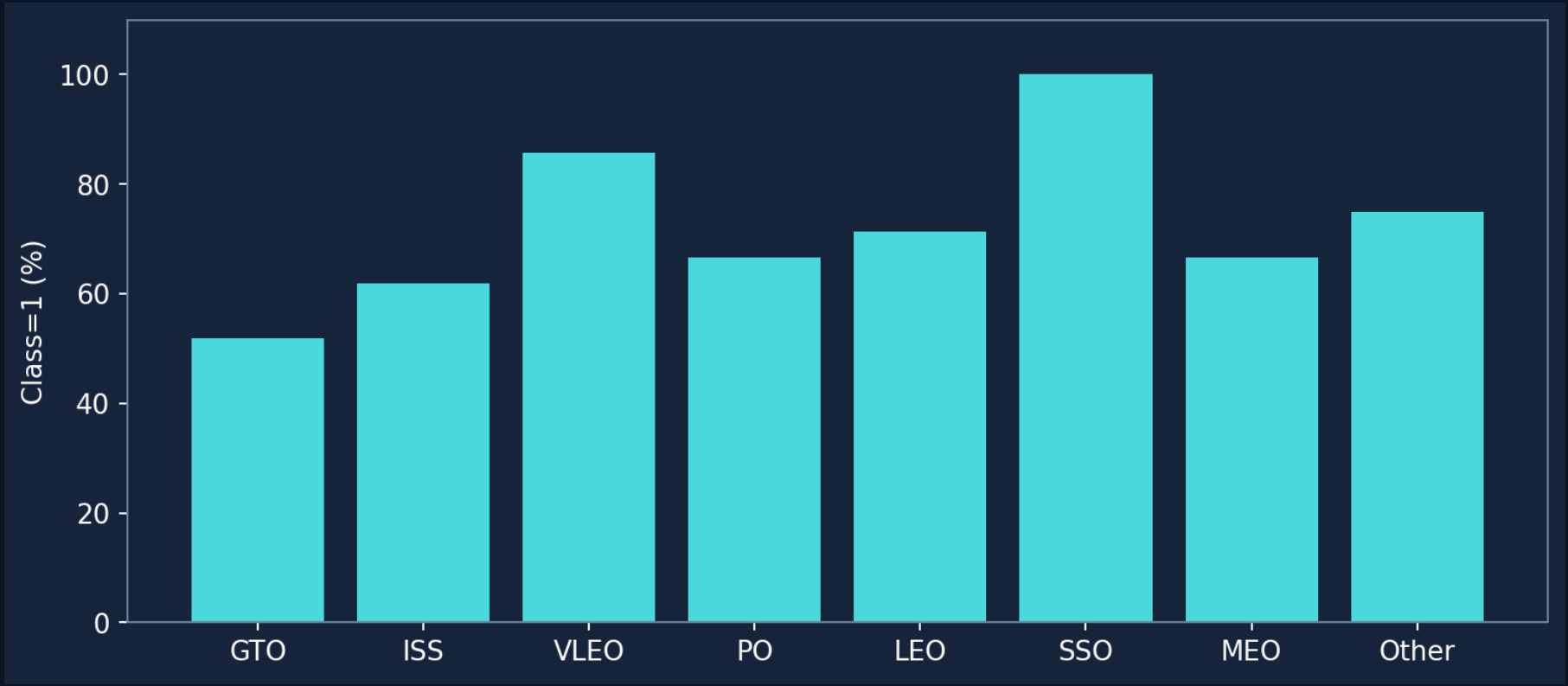


Chart: 90 total launches, including 19 records in 2020 and 18 each in 2017 / 2018.

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EDA results | orbit and class



Small orbit categories should not be over-interpreted; percentages have different denominators.

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EDA with SQL | analytical queries

Launch sites

```
SELECT LaunchSite, COUNT(*) AS launches FROM launches GROUP BY LaunchSite;
```

Payload by site

```
SELECT LaunchSite, SUM(PayloadMass) FROM launches GROUP BY LaunchSite;
```

Site recovery share

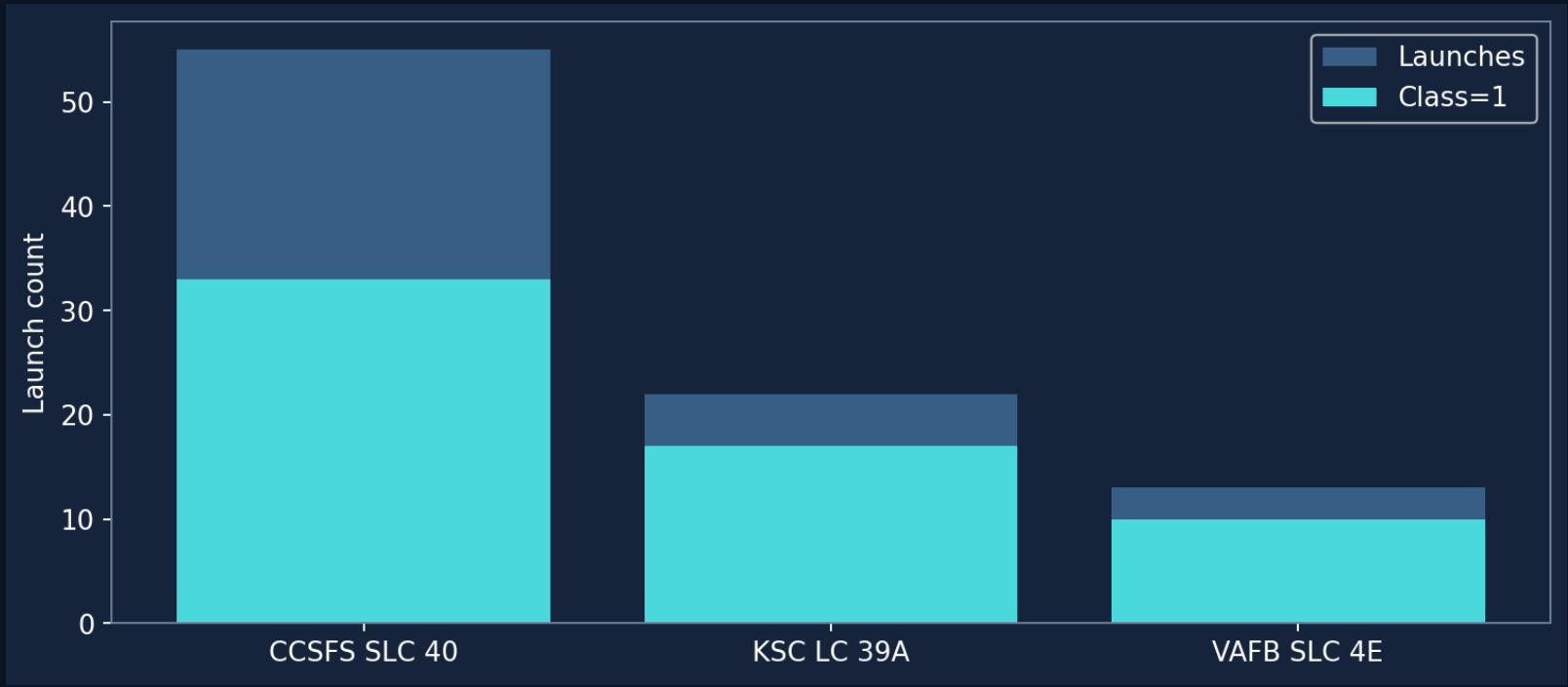
```
SELECT LaunchSite, AVG(Class)*100 FROM launches GROUP BY LaunchSite;
```

Annual trend

```
SELECT substr(Date,1,4) AS yr, COUNT(*) FROM launches GROUP BY yr;
```

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SQL results | launch sites and payload



CCSFS SLC 40

55 launches | 60.0% positive

KSC LC 39A

22 launches | 77.3% positive

VAFB SLC 4E

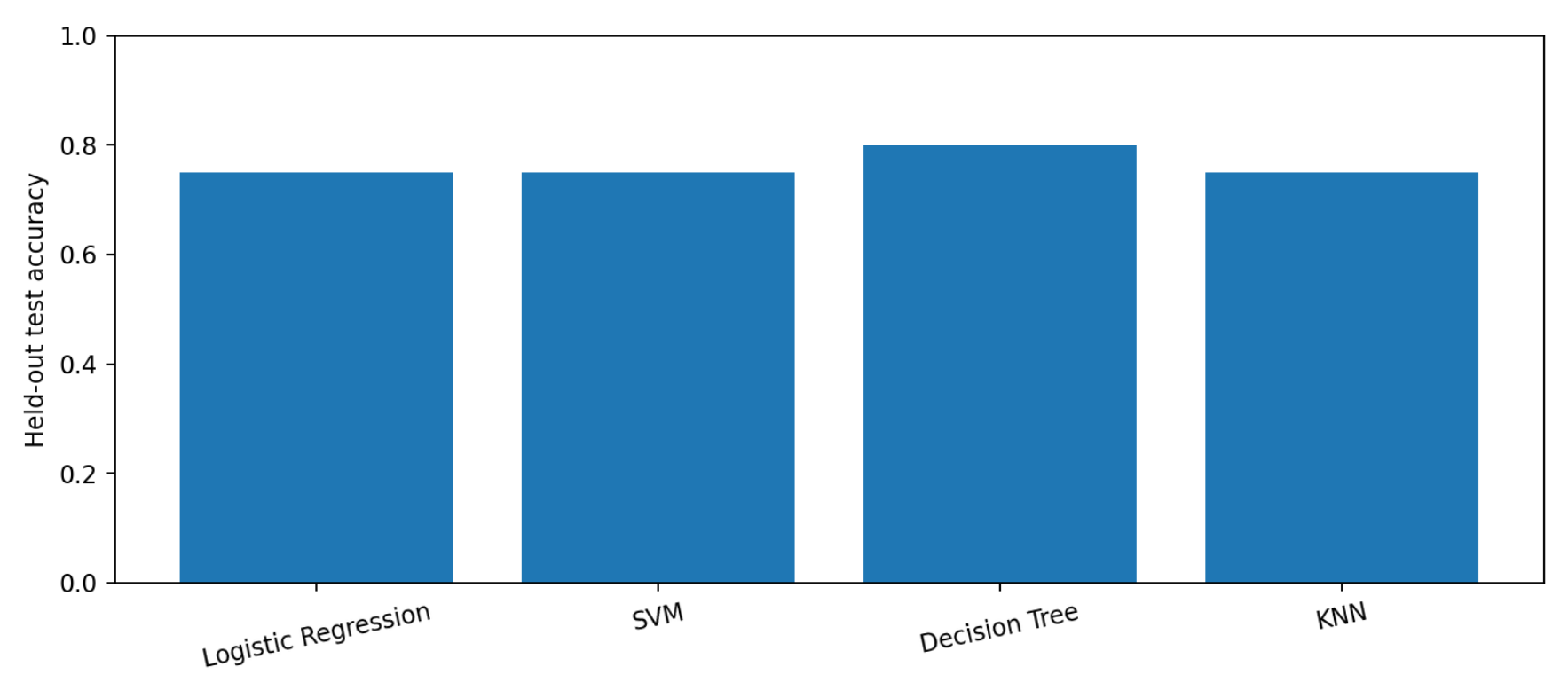
13 launches | 76.9% positive

Total payload (kg): CCSFS 305,987 | KSC 168,179 | VAFB 76,953 (rounded).

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Falcon 9 landing outcome | historic teaching sample

Predictive modeling | measured results

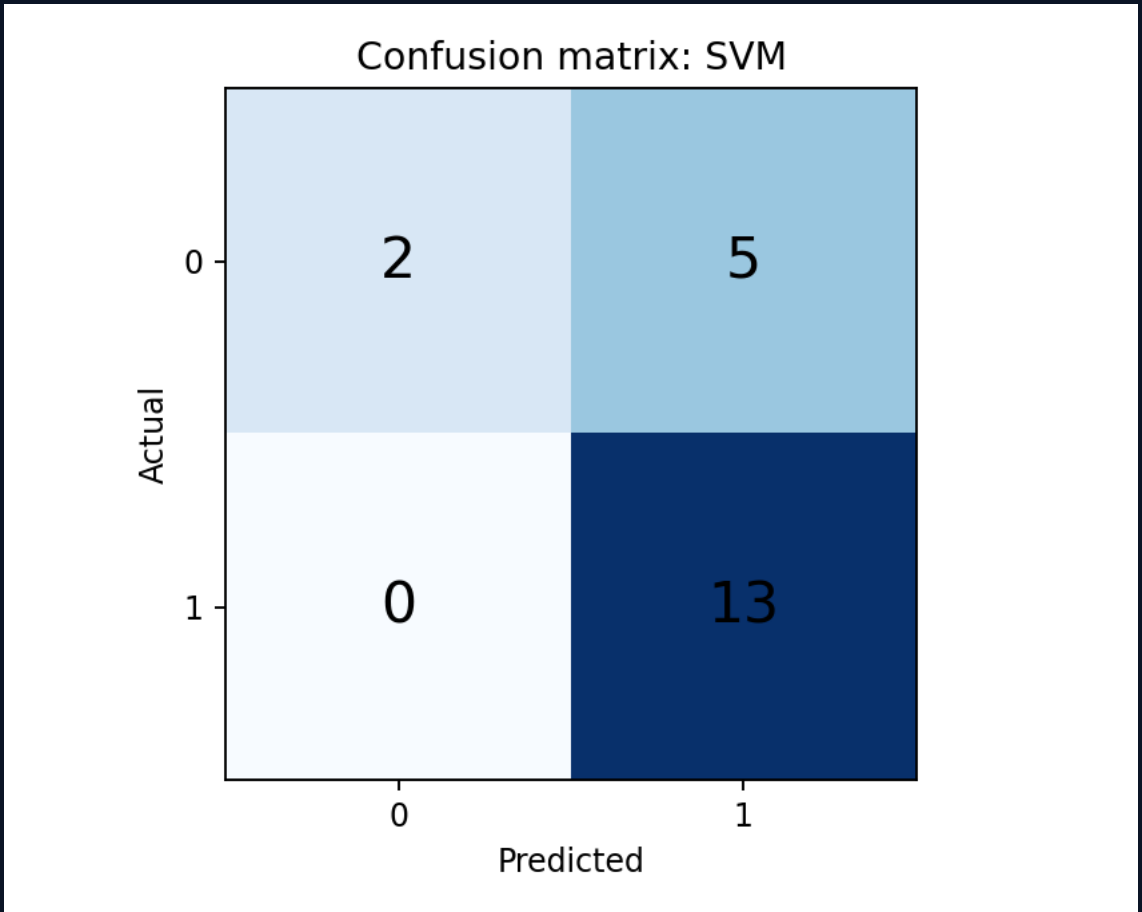


Test: 20 launches | train: 70 | stratified | 5-fold grid search within training data

Best TEST accuracy: Decision Tree (80%); SVM best cross-validation accuracy (90%).

Source: public 90-row course-derived historical dataset | Repository URL: pending publication

Model evaluation | confusion matrix



SVM — cross-validation leader

CV accuracy: 90%

Test accuracy: 75%

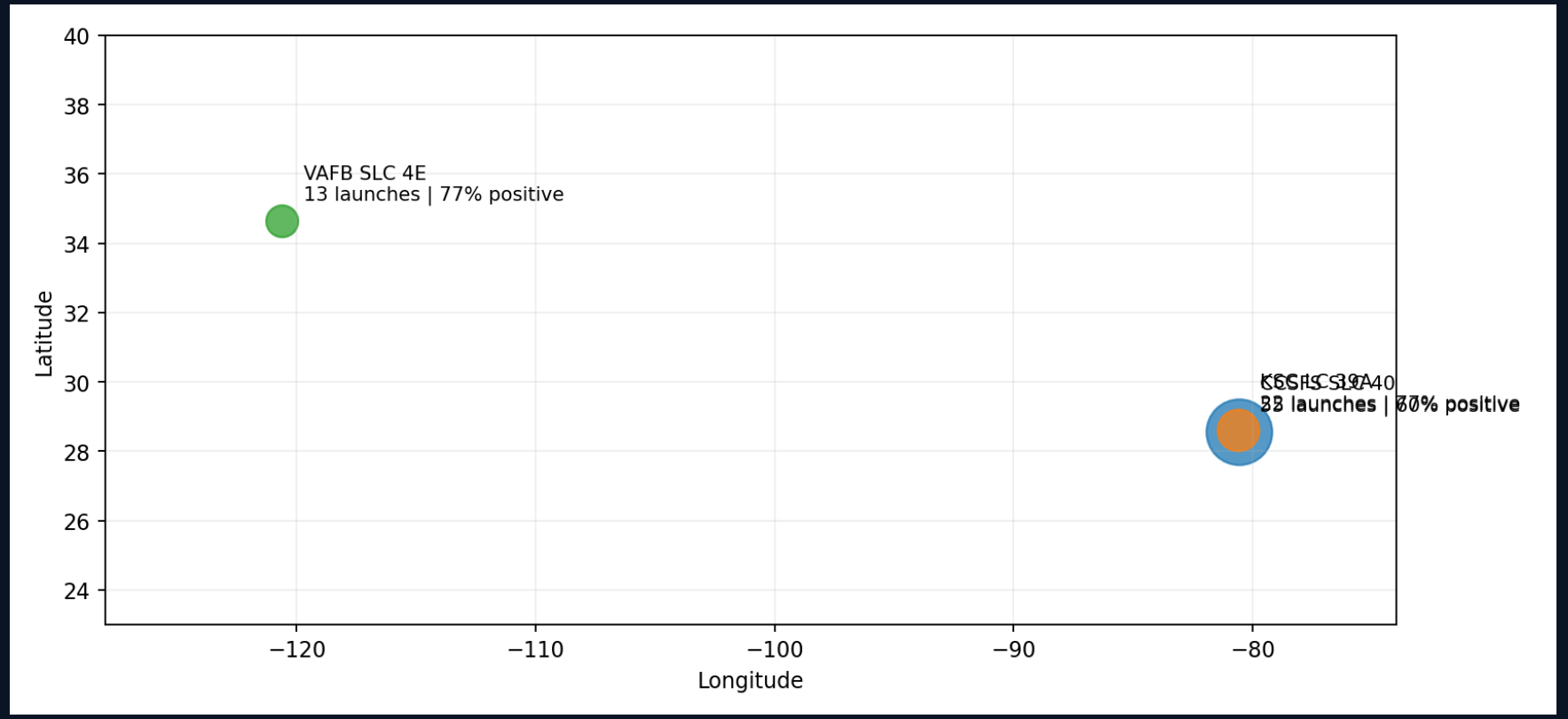
Decision Tree — test leader

Test accuracy: 80%

Precision / recall / F1: 0.846

Source: public 90-row course-derived historical dataset | Repository URL: pending publication

Folium results | launch-site marker evidence



Interactive map: launch_sites_map.html includes 3 clickable site markers and recovery share.

Coastline and highway distances not computed; they are not represented as measured results.

Source: public 90-row course-derived historical dataset | Repository URL: pending publication

Interactive Dash | source-backed visual analytics

Included runnable dashboard: dashboard.py

Dropdown filters all sites or a single launch site.

Pie chart displays positive recovery outcome percentage.

Scatter plot compares payload mass (kg) with outcome by orbit.

Run `python dashboard.py`, then visit localhost:8050.

A live rendered browser screenshot is not included as a verified artifact.

Source: public 90-row course-derived historical dataset | Repository URL: pending publication

Conclusions and business implications

Historical positive recovery: 60 of 90 Falcon 9 records (66.7%).

Launch sites: CCSFS 60.0%; KSC 77.3%; VAFB 76.9%.

Decision Tree: best held-out test accuracy, 80% (20 test rows).

SVM: best five-fold training CV accuracy, 90%; test accuracy 75%.

Recommendation: evaluate out-of-time data before operational decisions.

Caveat: historic prepared labels include positive ocean recovery outcomes.

Repository URL must be supplied after publication.

References and submission requirements

Public data: github.com/Abhigyan-RA/SpaceX-capstone

dataset_part_2.csv | 90 historic records | 2010-2020

Packaged: analysis notebook, SQL, 4 trained models, interactive map, Dash app.

No claim is made that live API / Wikipedia collection was executed.

Map site-level markers are measured; road/coast distances were not measured.

Required: publish source bundle to YOUR GitHub repository, add URL to slides.

Required: capture live dashboard if your grader demands UI screenshots.

REPOSITORY URL: [TO BE ADDED AFTER GITHUB PUBLICATION]